

## Part J: AI-Assisted Scientific Writing Comparison

Designing Verification Checkpoints to Mitigate Silent Errors in Agentic AI Pipelines

Name: Jahanvi Mishra | Section compared: Introduction, paragraph 2

### Version A — Original Paragraph

*The transition from language generation to agentic execution fundamentally changes the reliability problem. In a conventional software program, an incorrect computation may produce an exception, violate a type constraint, or trigger an explicit error-handling mechanism. An LLM-based agent, however, can produce an incorrect but syntactically valid tool argument, interpret an incorrect tool response as trustworthy, select an inappropriate next action, or generate a plausible explanation for an invalid intermediate state. These failures may not trigger any conventional exception. They are therefore silent errors: faults that remain operationally invisible while contaminating subsequent computation.*

### Version B — Prism-Assisted LaTeX Conversion

*However, increased autonomy also introduces new reliability challenges. In a traditional software system, incorrect behavior may result in explicit errors, exceptions, or failed tests. Agentic AI systems can instead produce incorrect intermediate reasoning, invalid tool calls, unsupported assumptions, or erroneous state transitions while continuing execution normally. Such failures may remain undetected because the system itself does not necessarily know that an intermediate result is incorrect.*

### Version C — Final Verified Version

*However, increased autonomy also introduces new reliability challenges. In a traditional software system, incorrect behavior may result in explicit errors, exceptions, or failed tests. Agentic AI systems can instead produce incorrect intermediate reasoning, invalid tool calls, unsupported assumptions, or erroneous state transitions while continuing execution normally. Such failures may remain undetected because the system itself does not necessarily know that an intermediate result is incorrect.*

(No manual wording change was required for this paragraph. It contains no equations, tables, figures, or citations, so it was not affected by the corrections made in Parts D, E, F, or I. Verification for this paragraph was therefore limited to confirming it compiles correctly and appears unchanged on the correct page of the final PDF.)

## Comparison and Analysis

### What Prism changed or reformatted:

Prism restructured the paragraph from a narrative, example-driven style into a more compact, declarative academic style suited to a LaTeX manuscript. The specific example list in the original (“a syntactically valid tool argument, interpret an incorrect tool response as trustworthy, select an inappropriate next action, or generate a plausible explanation for an invalid intermediate state”) was generalized into shorter phrasing (“incorrect intermediate reasoning, invalid tool calls, unsupported assumptions, or erroneous state transitions”). The two-sentence contrast between conventional software

and LLM-based agents was condensed into a single comparative sentence. The term “silent errors” itself was not restated in this particular paragraph, since it had already been defined earlier in the converted Introduction. No LaTeX-specific formatting (equations, citations, or cross-references) was required for this paragraph, since it contains none.

**What I manually corrected:**

I checked that the shortened, generalized wording still preserved the original's core technical claim — that agentic systems can fail in ways that do not raise a conventional software exception. I confirmed that no new claim, example, or scientific detail was introduced beyond what the original paragraph stated, and that the terms “explicit errors,” “exceptions,” and “failed tests” used in Version B are a fair paraphrase of the original's “exception,” “type constraint,” and “error-handling mechanism.” Since the meaning was preserved accurately, I did not need to rewrite this paragraph further, so Version C matches Version B for this particular passage.

**How I verified the final version:**

The full `main.tex` file, including this paragraph, was compiled using `pdflatex` and `bibtex` (three compilation passes) to confirm the document builds without errors. I then compared the compiled PDF's Introduction section side by side with the original paper to confirm no scientific claim, causal statement, or technical term had been altered, added, or removed in this paragraph during conversion. Because this paragraph does not reference any equation, table, figure, or citation, no cross-reference or bibliography check was applicable to it specifically; those checks were instead carried out on the relevant paragraphs elsewhere in the manuscript as documented in Parts D, E, F, G, and H.